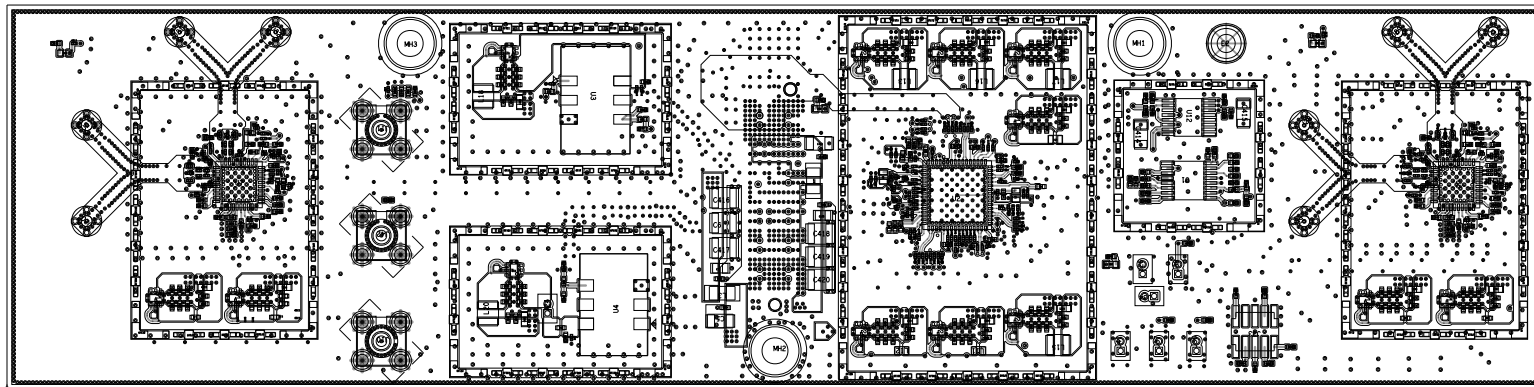




ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

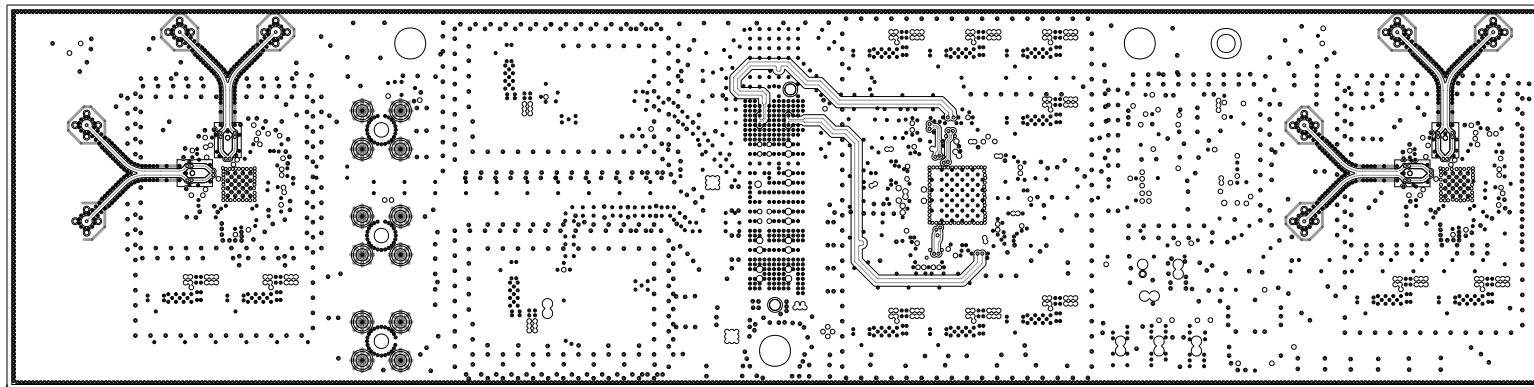
LAYER: L01_TOP

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 1 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

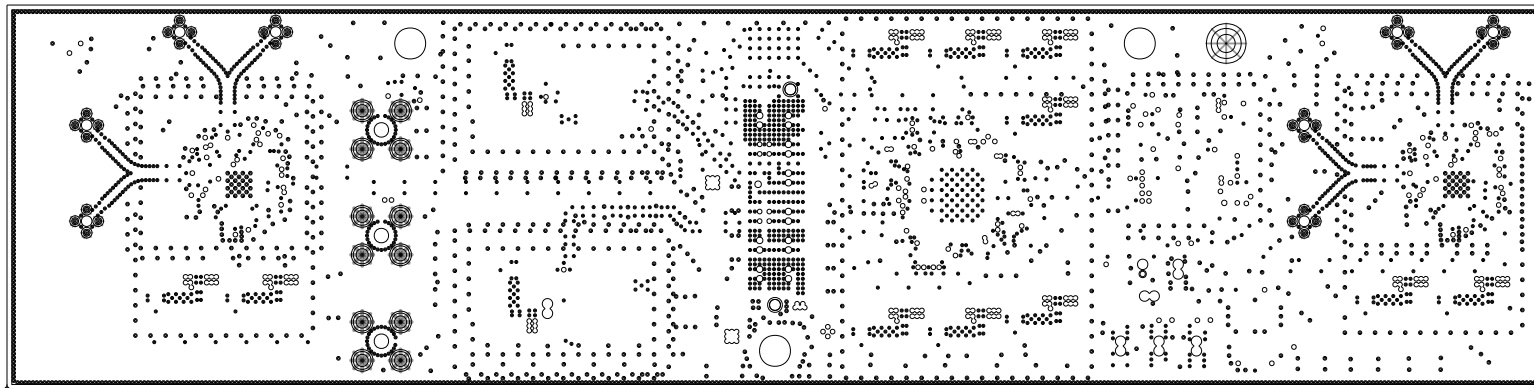
LAYER: L02_CLK/GND

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 2 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

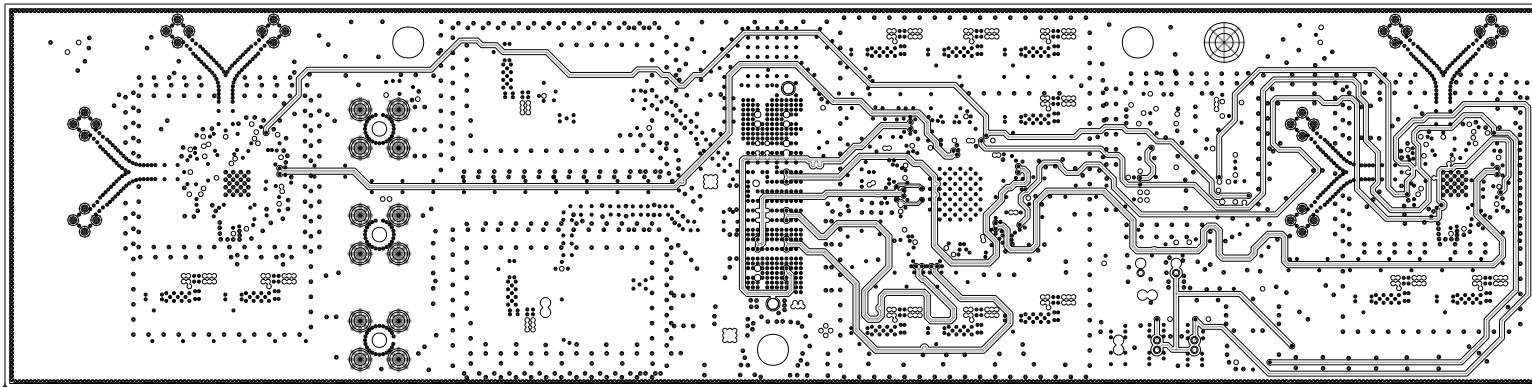
LAYER: L03_GND

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 3 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

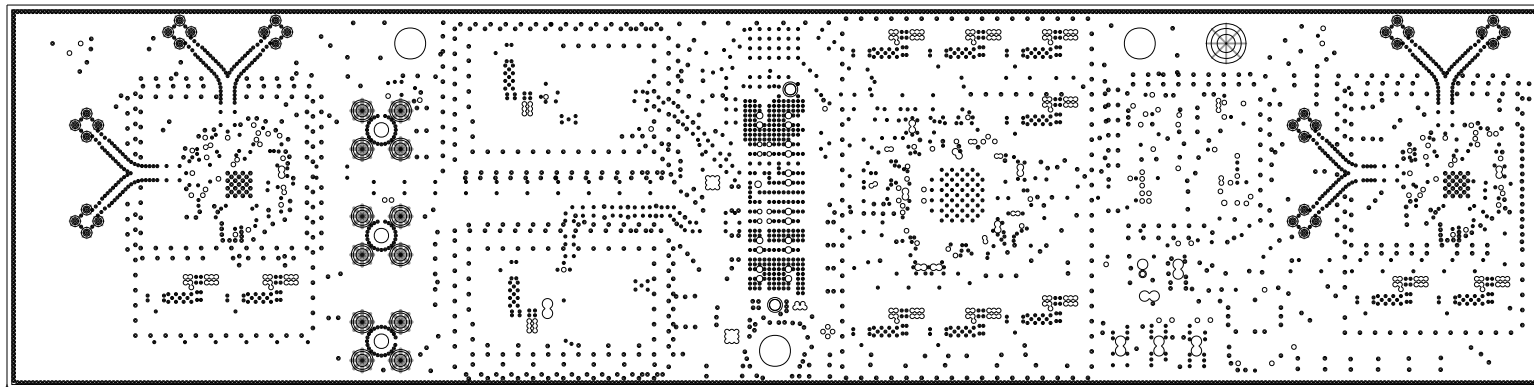
LAYER: L04_SIGNAL

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 4 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

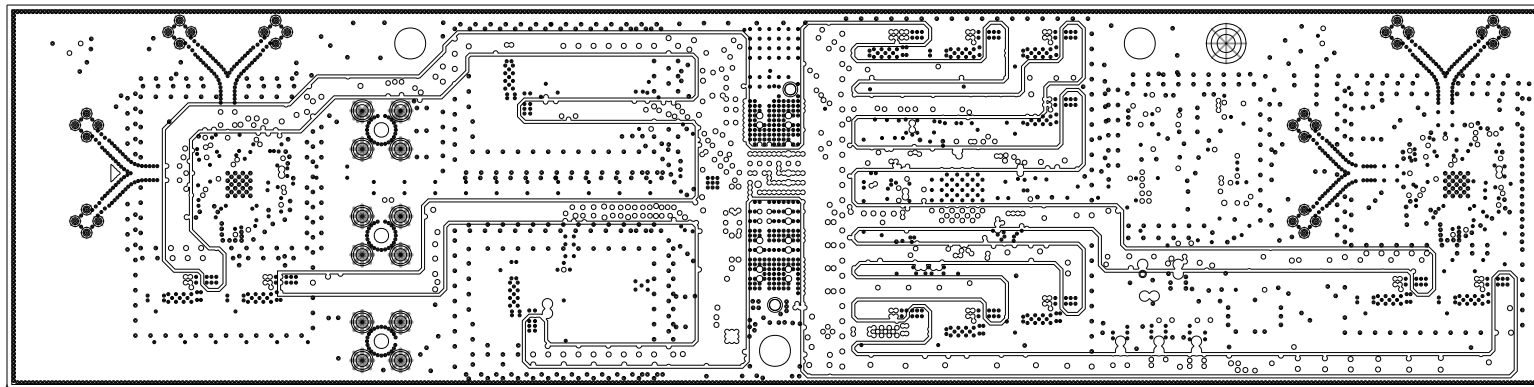
LAYER: L05_GND

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 5 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

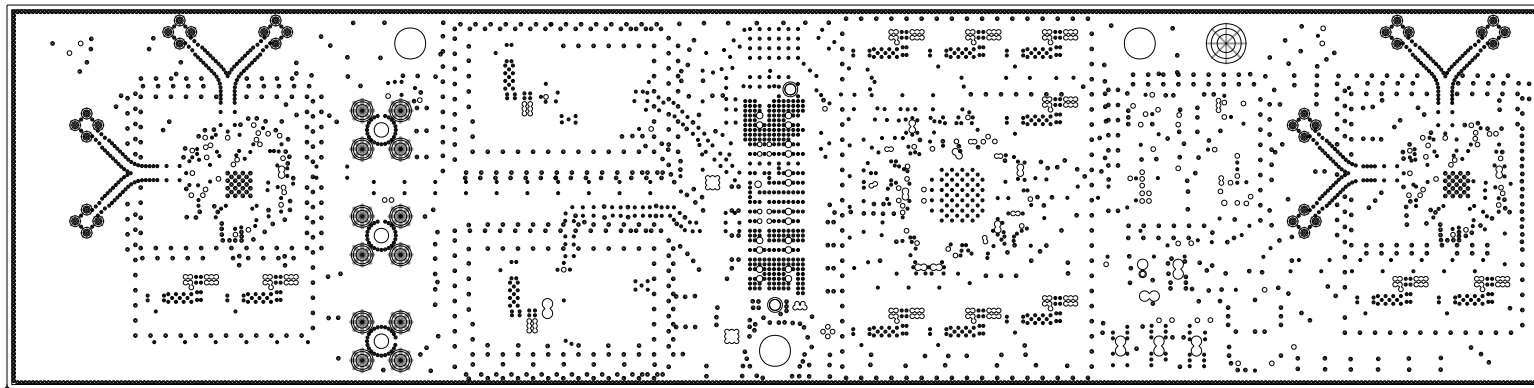
LAYER: L06_PWR

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 6 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

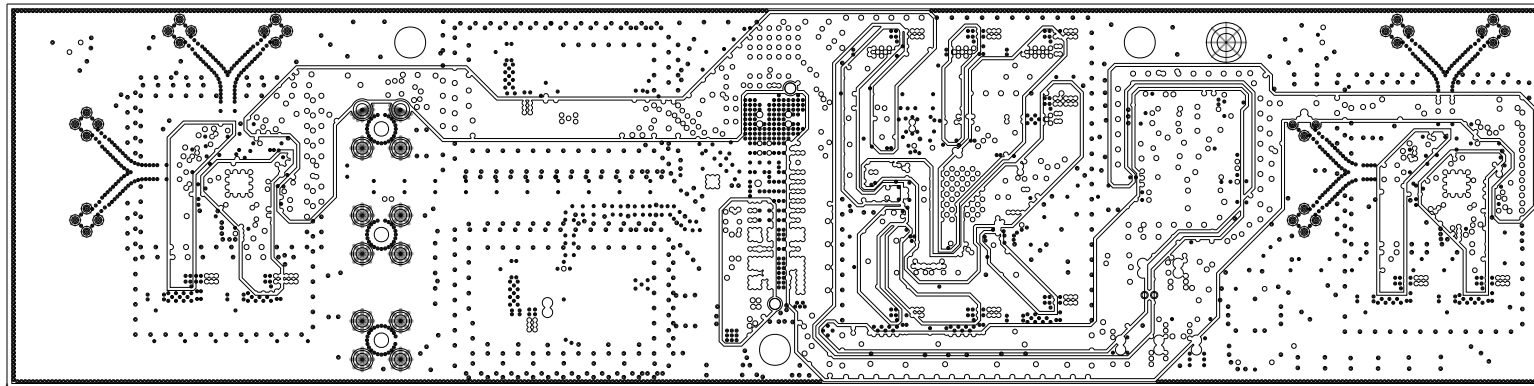
LAYER: L07_GND

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 7 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

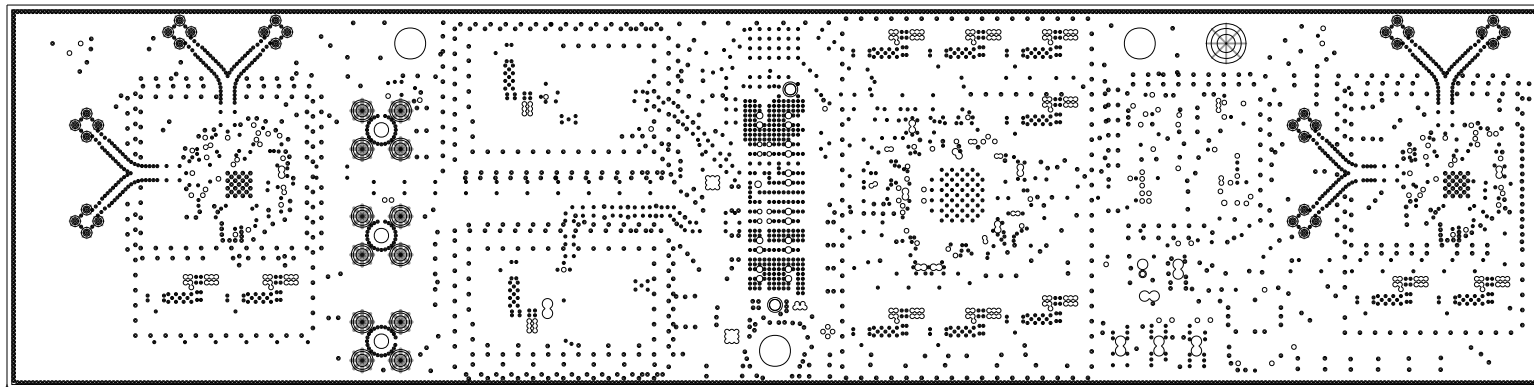
LAYER: L08_PWR

Designed by Xilinx

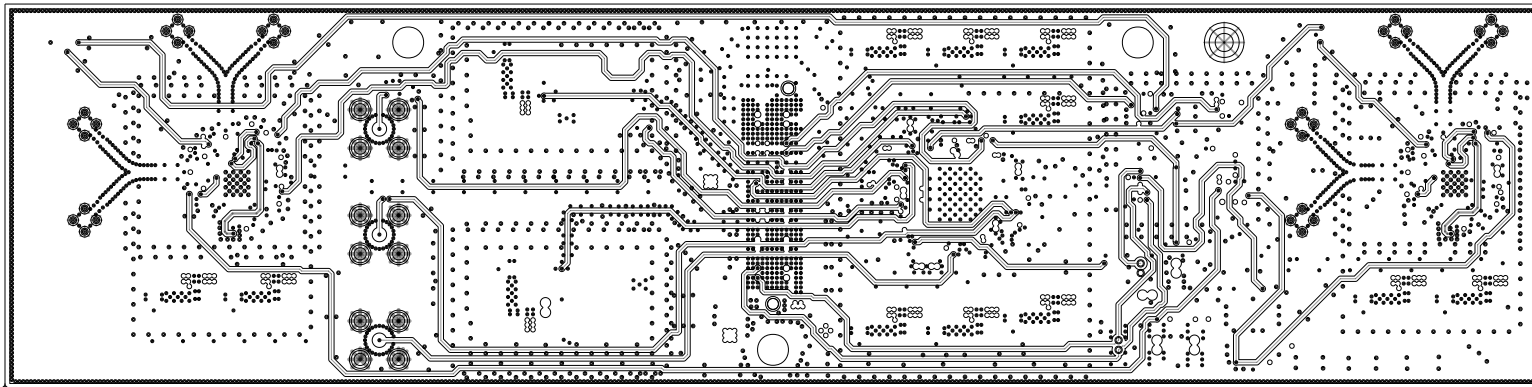
Layout designer: NHANO

DATE: 10-18-19

SHEET 8 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02		
PCB # 128-05007-01		LAYER: L09_GND
Designed by Xilinx		DATE: 10-18-19
Layout designer: NHANO		
SHEET 9 OF 18		



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

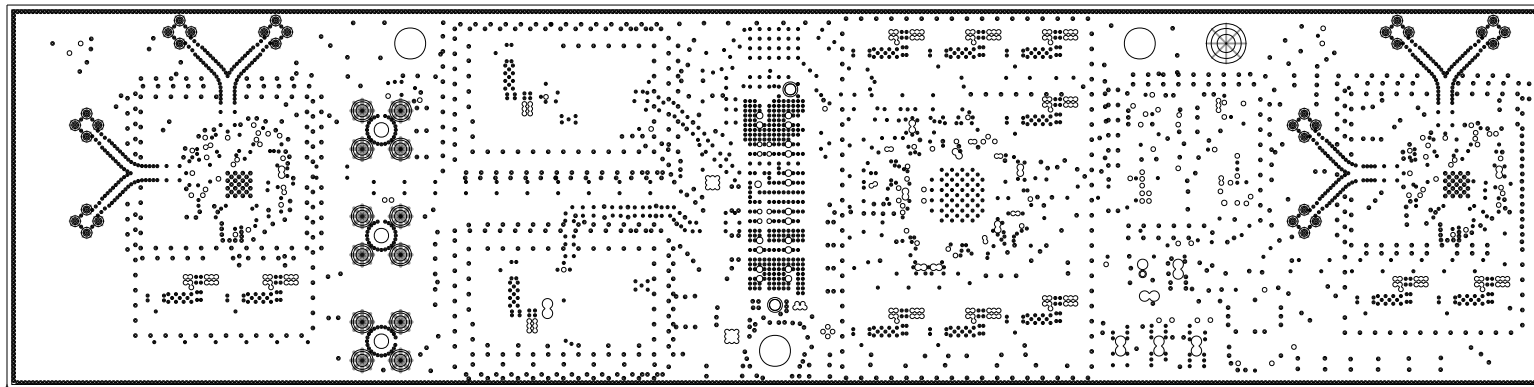
LAYER: L10_SIGNAL

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 10 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

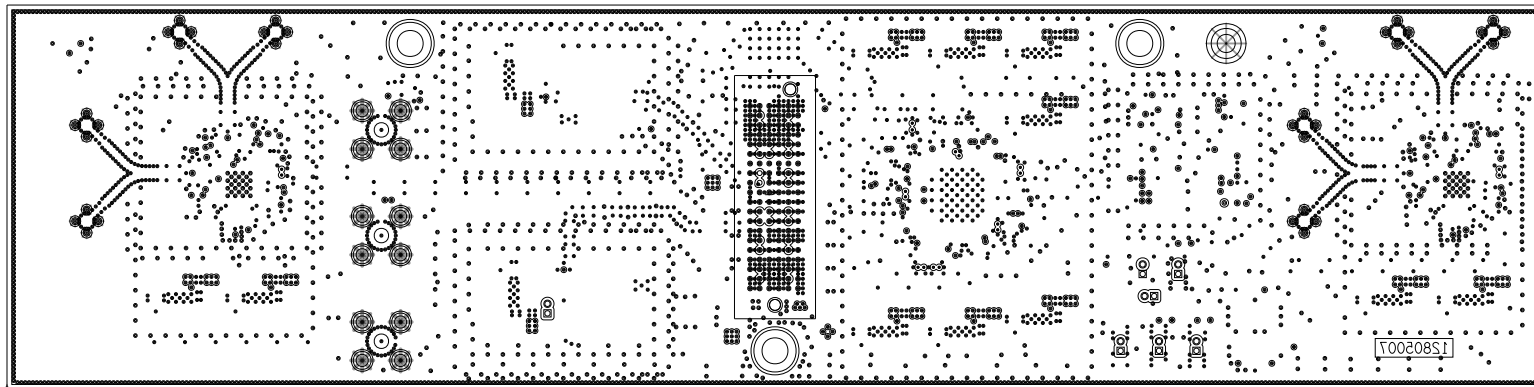
LAYER: L11_GND

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 11 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

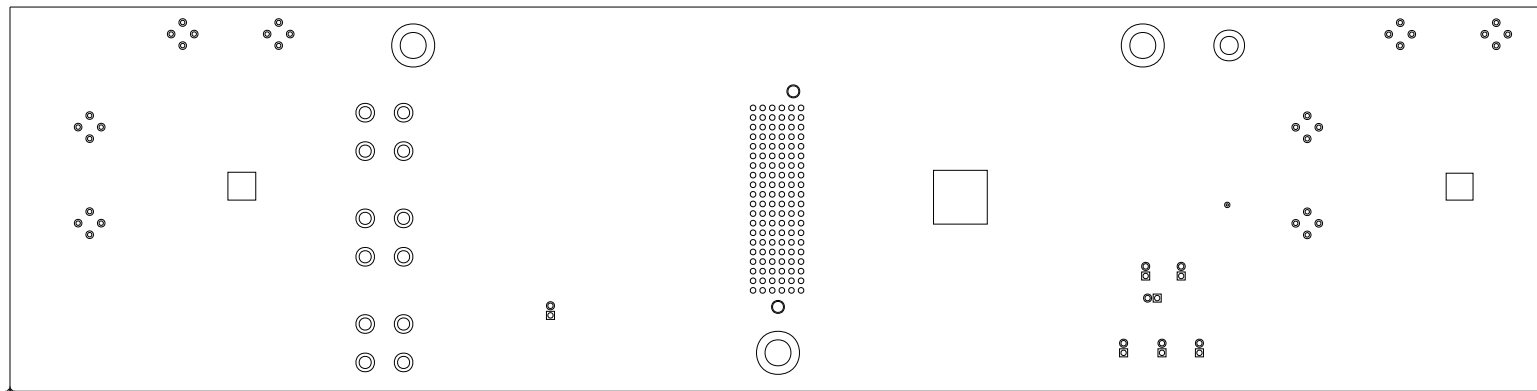
LAYER: L12_BOTTOM

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 12 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

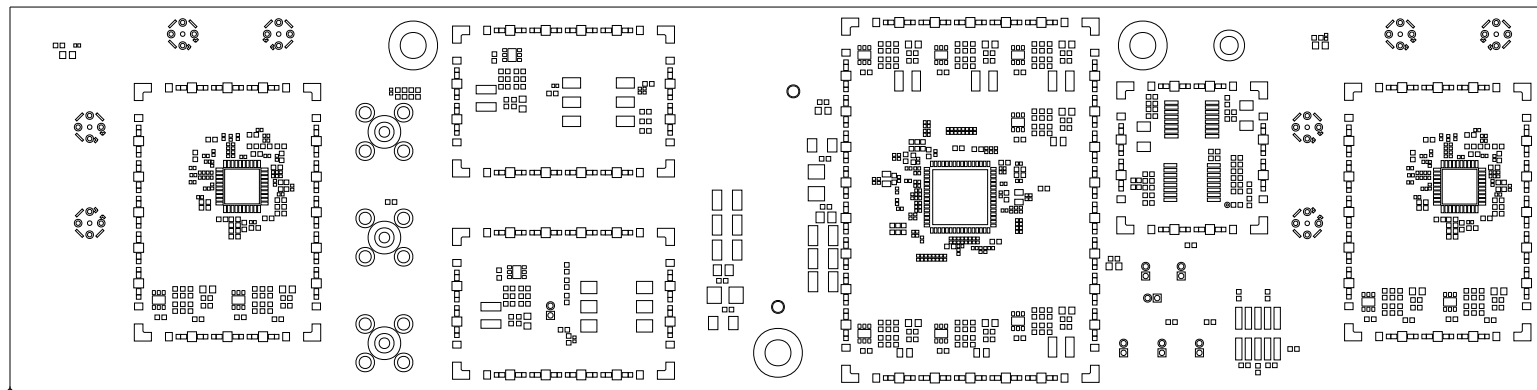
LAYER: MASK_BOTTOM

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 13 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

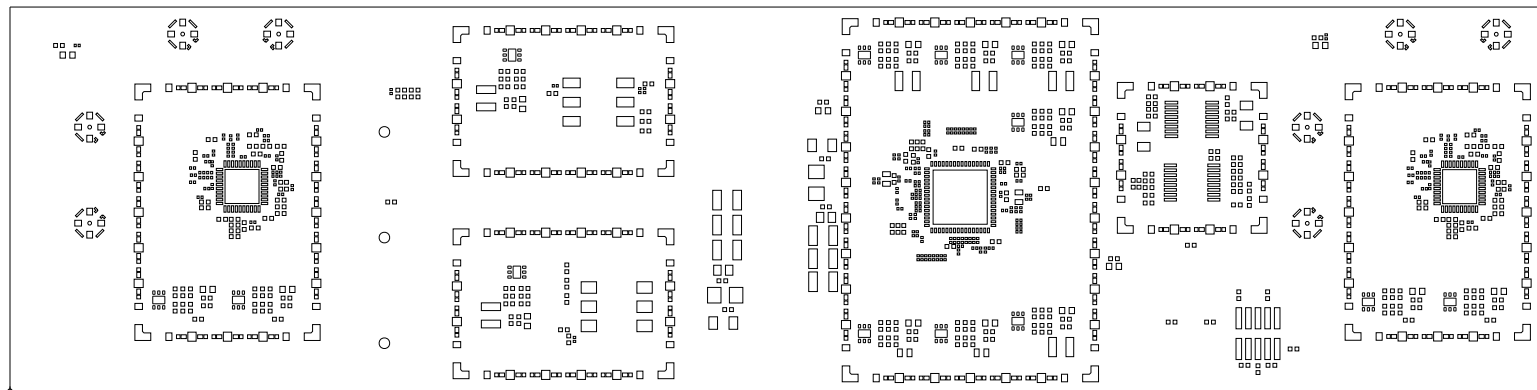
LAYER: MASK_TOP

Designed by Xilinx

Layout designer: NHANO

DATE: 10-18-19

SHEET 14 OF 18



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02

PCB # 128-05007-01

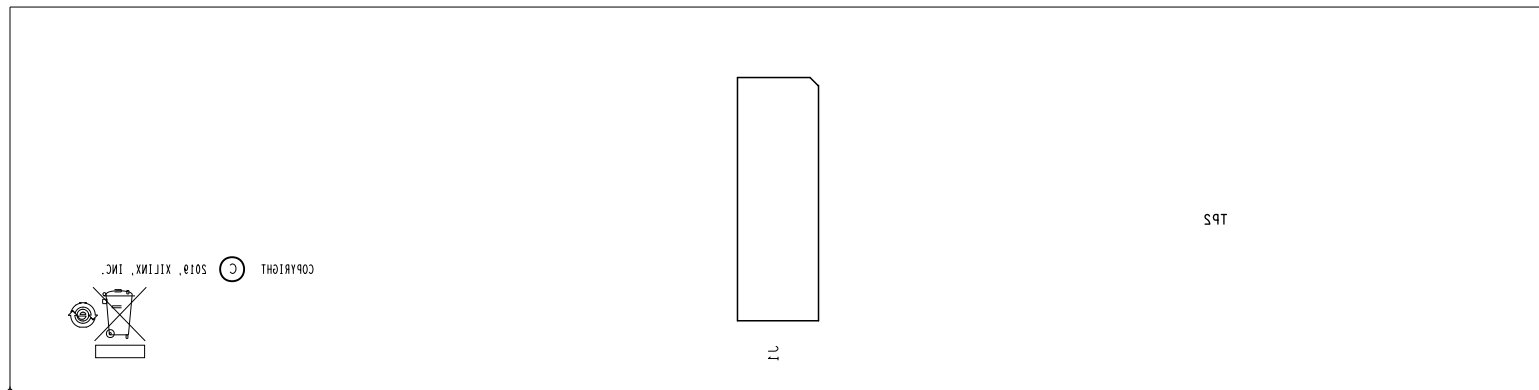
LAYER: PASTE_TOP

Designed by Xilinx

Layout designer: NHANO

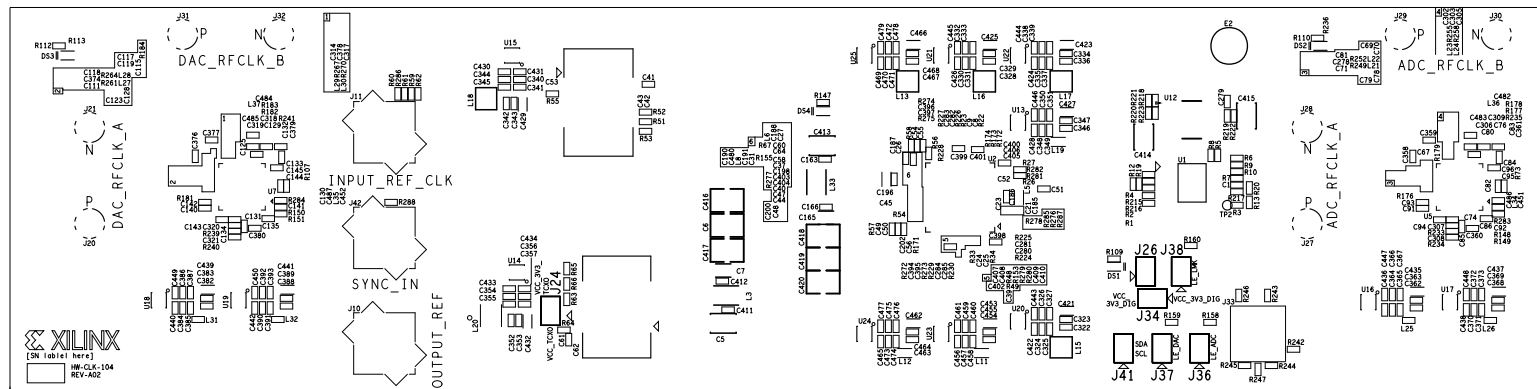
DATE: 10-18-19

SHEET 16 OF 18



S9T

ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02		
PCB # 128-05007-01	LAYER: SILK BOTTOM	
Designed by Xilinx Layout designer: NHANO		DATE: 10-18-19
SHEET 17 OF 18		



ARTWORK, ROHS COMPLIANT, HW-CLK-104 REV-A02	
PCB # 128-05007-01	LAYER: TOP SILK
Designed by Xilinx Layout designer: NHANO	DATE: 10-18-19
SHEET 18 OF 18	

DRILL CHART: TOP to L02_CLK				
ALL UNITS ARE IN MILS				
FIGURE	FINISHED_SIZE	TOLERANCE_DRILL	PLATED	QTY
.	5.0	+0.0/-5.0	PLATED	182

DRILL CHART: TOP to BOTTOM				
ALL UNITS ARE IN MILS				
FIGURE	FINISHED_SIZE	TOLERANCE_DRILL	PLATED	QTY
.	8.0	+0.0/-10.0	PLATED	1971
.	10.0	+0.0/-10.0	PLATED	2152
.	12.0	+2.0/-2.0	PLATED	1
•	23.0	+3.0/-3.0	PLATED	32
•	28.0	+3.0/-3.0	PLATED	7
•	28.0	+3.0/-3.0	PLATED	7
#	63.0	+3.0/-3.0	PLATED	12
Ⓢ	94.0	+3.0/-3.0	PLATED	1
Ⓢ	57.0	+2.0/-2.0	NON-PLATED	2
135	135.0	+3.0/-3.0	NON-PLATED	3

NOTES:

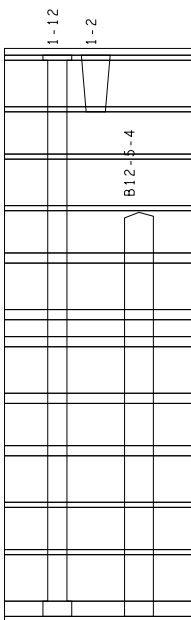
- BOARD SHALL MEET THE INSPECTION CRITERIA OF IPC-A-600, CLASS 2, CURRENT REVISION.
- MATERIAL: REFER TO STACK-UP. THICKNESS .068" +/- 10%
- WEIGHT OF ALL COPPER LAYERS SHALL NOT BE LESS 0.5 OZ. PER SQUARE FOOT.
- IMPEDANCE REQUIREMENTS REFER TO CHART. TOLERANCE +/-10% EXCEPT AS NOTED.
- APPLY SOLDER MASK OVER BARE COPPER(SMBOC) IAW IPC-SM-840, BOTH SIDES, USING LPI, COLOR GREEN
- FINISH: GOLD IMMERSION.
- SILKSCREEN TOP SIDE/BOTH SIDES WITH NON-CONDUCTIVE EPOXY BASED INK. COLOR SHALL BE WHITE A CONTRASTING INK WITH RESPECT TO SOLDER MASK COLOR. DISTORTION OF SILKSCREEN IS ACCEPTABLE OVER TRACES. EPOXY INK ON PLATED LANDS IS NOT ACCEPTABLE.
- VENDOR LOGO AND DATE CODE TO BE MARKED FAR SIDE. MAXIMUM HEIGHT .12 INCHES.
- 100% ELECTRICAL TEST REQUIRED FOR CONTINUITY. BOARD SHALL HAVE A UL RATING OF 94V-0. UL SYMBOL AND RATING SHAL BE MARKED FAR SIDE.
- REMOVE ALL UNUSED PADS FROM INTERNAL LAYERS.
- SOLDER MASK FOLLOW GERBER DATA FOR TOP & BOTTOM SIDE.

REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
A	PROTOTYPE	07-09-18	

BACKDRILL: BOTTOM to L05_GND					
ALL UNITS ARE IN MILS					
FIGURE	BD_SIZE	MNC_LAYER	MAX_DEPTH	MFG_STUB	QTY
.	18.0	L04_SIGNAL	48.0	-	18

NOTES:

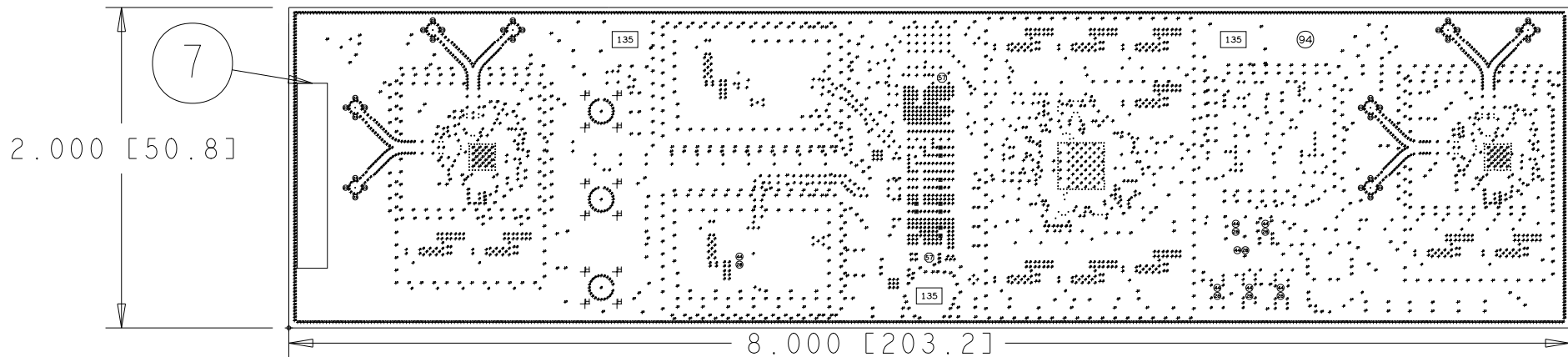
- MNC_LAYER: MUST-NOT-CUT-LAYER
- MAX_DEPTH: DEPTH FROM START LAYER TO THE SURFACE OF MUST-NOT-CUT-LAYER
- MFG_STUB : MANUFACTURING STUB LENGTH

	* SURFACE - AIR 0 MIL
	* DIELECTRIC - CONFORMAL COAT 0.8 MIL
	L1 TOP PLANE - 050Z_CU 0.6 MIL
	* DIELECTRIC - 001-ITERA-MT40-2X1067-RC76-PREG 5.5 MIL
	L2 L02_CLK PLANE - 0.6 MIL
	* DIELECTRIC - 002-ITERA-MT40-2X1067-RC72-5CORE 5 MIL
	L3 L03_GND PLANE - 050Z_CU 0.6 MIL
	* DIELECTRIC - 001-ITERA-MT40-2X1067-RC76-PREG 5.5 MIL
	L4 L04_SIGNAL CONDUCTOR - 0.6 MIL
	* DIELECTRIC - 002-ITERA-MT40-2X1067-RC72-5CORE 5 MIL
	L5 L05_GND PLANE - 10Z_CU 1.2 MIL
	* DIELECTRIC - 001-ITERA-MT40-2X1067-RC76-PREG 5.5 MIL
	L6 L06_PWR PLANE - 10Z_CU 1.2 MIL
	* DIELECTRIC - 003-ITERA-MT40-1X1067-RC68-2CORE 2 MIL
	L7 L07_GND PLANE - 1.2 MIL
	* DIELECTRIC - 001-ITERA-MT40-2X1067-RC76-PREG 5.5 MIL
	L8 L08_PWR PLANE - 10Z_VLP2_CU 1.2 MIL
	* DIELECTRIC - 002-ITERA-MT40-2X1067-RC72-5CORE 5 MIL
	L9 L09_GND PLANE - 10Z_CU 1.2 MIL
	* DIELECTRIC - 001-ITERA-MT40-2X1067-RC76-PREG 5.5 MIL
	L10 L10_SIGNAL CONDUCTOR - 050Z_CU 0.6 MIL
	* DIELECTRIC - 002-ITERA-MT40-2X1067-RC72-5CORE 5 MIL
	L11 L11_GND PLANE - 050Z_VLP2_CU 0.6 MIL
	* DIELECTRIC - 001-ITERA-MT40-2X1067-RC76-PREG 5.5 MIL
	L12 BOTTOM CONDUCTOR - 050Z_HTE_CU 10Z PLATING 1.8 MIL
	* DIELECTRIC - CONFORMAL COAT 0.8 MIL
	* SURFACE - AIR 0 MIL

DESIGN CROSS SECTION CHART
TOTAL THICKNESS 68 MIL

LINE WIDTH IMPEDANCE CHART FOR REFERENCE

LAYER	SINGLE ENDED LINE WIDTH	DIFF PAIR LINE WIDTH	DIFF PAIR LINE SPACE	SINGLE 50 OHMS	DIFF 100 OHMS
1	12.50 MILS	7.30 MILS	4.00 MILS	+/- 10%	+/- 10%
2	5.70 MILS	5.80 MILS	15.00 MILS	+/- 10%	+/- 5%
4	5.70 MILS	4.00 MILS	3.75 MILS	+/- 10%	+/- 5%
10	5.70 MILS			+/- 10%	



- SOLDER MASK REGISTRATION TO BE WITHIN DIAMETRICAL TRUE POSITION OF + / - 0.002 WITH APPLICABLE HOLE / PAD.
- 274X GERBERS/ODB++ USED FOR FAB MUST BE VERIFIED AGAINST THE PROVIDED IPC NETLIST.
- USE ARTMASTER # 128-05007-01
- DRILL SIZES IN CHART ARE CONSIDERED FINISHED SIZE.
- 2um SURFACE ROUGHNESS COPPER ON ALL LAYERS.
- ALL VIA IN PAD TO BE COMPLETELY FILLED, PLANARAIZED SMOOTH AND PLATED OVER ON SURFACE. USE SanEi NON-CONDUCTIVE EPOXY OR EQUIVILANT. MINIMUM OF .0007 TOP BE PLATED ON SURFACE. VIA-IN-PAD MUST INCLUDE WRAP REQUIREMENTS PER IPC 60128
- BUILD BOARDS IN A PANEL (MINIMUM 4 BOARD) BEST FIT ARRAY. PROVIDE ARRAY GERBER FOR ASSEMBLY. ONCE YOU HAVE THE ARRAY. PLEASE SEND TO XILINX.
- BOARD MUST BE RoHS COMPLIANT. VENDOR TO SUPPLY RoHS COMPLIANCY WITH BOARD.
- BAKE BOARDS AT 225 DEGRESS FOR 4 HOURS TO REMOVE ANY MOISTURE AFTER FABRICAITON.

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ON: 2 PL DECIMALS +/- .010 3 PL DECIMALS +/- .005 ANGLES + FRACTIONS +	SIGNATURES		DATE	 2100 LOGIC DR. SAN JOSE, CA 95124 HWCP - HARDWARE & CONFIGURATION PLATFORMS HW-CLK-104A ANALOG CLOCK MODULE FABRICATION			
	DRAWN NHANO		10-18-19				
	CHECKED						
	ENGRG JIN WU		10-18-19				
	ISSUED						
				SIZE D	FSCM NO	DWG NO 128-05007-01	REV:A02
				SCALE NONE			SHEET